Team Hack2Publish — Research Effort Summary

ARC White-Box Estimation Challenge 2026, Phase 2. Team: mdhamidhosen, esfer_sami, foyshal_sk. Status as of 21 October 2026.

Counted from the repositories and experiment ledgers, the team ran **more than 500 experiments**. Because the counts come from files, a few experiments may be counted in two places; the overall scale is correct.

Architectures and new methods

- **More than 60** distinct architecture and representation ideas were tested to a decision.
- **20 consultation rounds** with outside AI systems (GPT, Astra, Gemini, Grok, DeepSeek and others; plus internal Claude subagents); every proposal was measured before use.
- The kill register lists **about 80** measured and closed avenues.

Architectures that survived and were submitted:

Architecture	What it does	Graded effect
V31	Spherical-input identity (exact, zero FLOPs)	raw −1.1 %
V55	κ_4-diagonal pattern corrector of the mean closure	−5.4 %
V56	Hermite cubic-star feature for the κ_4 pattern	−4.4 %
E-series calibration (mdhamidhosen)	Gauss-Newton refit of 24 calibration groups and the κ_4 corrector	−3.9 %
e30_a + κ_4 corrector	Our corrector on another team's public (MIT) chain	−2.1 % vs e30_a
e299safe	Fixed-basis old level + C_pre compaction in our own lineage	−2.4 %
Refitted table (N2 → N4a)	e30_a's correction table refitted with our non-local features	−6.3 %

Experiments and measurements

Item	Count
Probe and measurement scripts (V-series)	130
Estimator variants (with knobs or oracles)	50
Result files	280
100-MLP Kaggle harness runs (result folders)	159
Kaggle kernels created	132
E-series ledger (mdhamidhosen)	133 experiments, 117 estimator versions
Planned-experiment queue	62

Grader submissions

- About **134 graded submissions** in total for the team (board count, 20 October): about 66 E-series submissions by mdhamidhosen and the rest from the V-series and the final-days line.
- Score improved from **5.418e-9 to 2.7858e-9 (−48.6 %)**.
- Best public rank was **24** (26 September); the rank on 20 October is **34** (displayed 2.8e-9).

Writing and verified results

- Main write-up: 12 pages. Full technical report (appendix): 56 pages.
- Four results verified numerically: the parity theorem, the proof of the $\sqrt{(6/(r+8))}$ floor, the sign of the Gaussian star, and the direct measurement of channel-error cancellation.